

Classwork: TypeScript Decision Arcade

100 points + 10 bonus | Use typescriptlang.org/play during class

Student name

Date

Part 1 — Predict Before Run

15 points

Read each line. Predict the output before you click Run in the TypeScript Playground.

```
const player: string = "Nova";  
let energy: number = 40;  
energy = energy + 15;  
console.log(player);  
console.log(energy);  
console.log(energy >= 50);
```

Line 1 output

Line 2 output

Line 3 output

Did Run match?

Part 2 — Rebuild an if / else Gate

15 points

Fuel 50 or higher launches. A lower value must refuel. Complete the missing condition and messages.

Condition inside if (...)

True-branch message

Else-branch message

Test fuel = 50. What prints, and why?

Rank Engine and Boundary Bosses

Classwork continued

Part 3 — Build the Rank Engine

20 points

Use: 90+ Legendary, 70+ Hero, 50+ Explorer, and anything lower Rookie. Write one connected ladder.

Your TypeScript code:

Part 4 — Boundary Boss Battle

15 points

Predict first, then run each value. Record the output you actually see.

Score 90

Score 89

Score 70

Score 69

Score 50

Score 49

Why test a boundary and the number just below it?

AI Bug Bounty

Classwork continued

Part 5 — Find and Repair the Logic Bug

15 points

An AI assistant suggested this code. It compiles, but its condition order causes a wrong answer.

```
const score: number = 95;

if (score >= 50) {
  console.log("Explorer");
} else if (score >= 70) {
  console.log("Hero");
} else if (score >= 90) {
  console.log("Legendary");
} else {
  console.log("Rookie");
}
```

1. What prints for 95?

2. What should print?

3. Why does the wrong branch win?

4. Rewrite only the three conditions in the correct order:

5. Why must a human test AI-generated code?

Design Your Decision Game

Classwork conclusion and bonus

Part 6 — Create a Tiny Game

20 points

Use one number variable, one if, at least two else if branches, one final else, and four different messages.

Game theme or title

Your TypeScript code:

Test value 1

Output

Boundary test

Output

Bonus — Explain a Full Trace

+10 points

Choose one value. Explain each false check, the first true check, and why the ladder stops.

Submit to Microsoft Teams AND Ishwari Raut ma'am.